

GTOS Handbook - Master Overview

GTOS Enterprise Engineering Handbook - Final Edition 1.0

Source Chapters

- GTOS Completion Definition
- Master Table of Contents
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- GTOS Enterprise Engineering Handbook
- Release Notes - Final Edition 1.0
- Version
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- Architectural Principles
- Target Operating Model
- Domain Map and Ownership
- End-to-End Global Trade Lifecycle
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- Data Architecture
- API, Event, and Tool Architecture
- Workflow, Saga, and Recovery Architecture
- AI-Native Architecture
- Security, Privacy, and Compliance
- Reliability, SRE, and Operations
- Engineering Standards
- Testing and Conformance
- Implementation Roadmap
- Domain Implementation Template
- Production Readiness Standard

GTOS Completion Definition

A GTOS capability is complete only when every required question below has an evidence-backed answer.

Business and ownership

- What business outcome does the capability produce?
- Which bounded context owns each proposition?
- Which external authority owns legal, financial, quality, carrier, or regulatory outcomes?
- Who may decide, approve, execute, correct, revoke, and appeal?

Domain model

- What are the canonical subjects and stable identities?
- What are the aggregates, entities, value objects, relationships, and invariants?
- What state machines exist and what guards every transition?
- What history, correction, supersession, contradiction, and dispute semantics apply?

Contracts and data

- What Commands, Events, queries, APIs, webhooks, files, and tools exist?
- Which service owns each database, schema, table, index, and migration?
- How are idempotency, concurrency, ordering, replay, reconciliation, and versioning handled?
- How are lineage, classification, retention, residency, and deletion handled?

Experience and automation

- Which personas and Portals participate?
- How are partial, stale, denied, failed, disputed, and outcome-unknown states displayed?
- What AI tasks are allowed, what autonomy level applies, and what human approval is required?
- What evidence and explanation accompany automated recommendations and actions?

Quality and operations

- What security and privacy controls are enforced?
- What SLOs, dashboards, alerts, runbooks, capacity limits, and recovery objectives exist?
- What unit, property, integration, contract, security, performance, chaos, migration, E2E, and DR tests prove the behavior?
- What release evidence demonstrates deployability and operability?

A module name, route, table, mock screen, plan, or generated document is never sufficient evidence by itself.

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- [Regulatory — Ubiquitous Language, Aggregates, and State](#) - 1,991 words
- [Regulatory — Commands, Events, APIs, Workflows, and Tools](#) - 1,995 words
- [Regulatory — Data, Tables, Migrations, Lineage, and Reconciliation](#) - 1,995 words
- [Regulatory — Portals, AI, Security, SLOs, Tests, and Operations](#) - 1,993 words
- [Production Fulfilment — Orientation and Evidence Baseline](#) - 1,997 words
- [Production Fulfilment — Ubiquitous Language, Aggregates, and State](#) - 1,996 words
- [Production Fulfilment — Commands, Events, APIs, Workflows, and Tools](#) - 2,000 words
- [Production Fulfilment — Data, Tables, Migrations, Lineage, and Reconciliation](#) - 2,000 words
- [Production Fulfilment — Portals, AI, Security, SLOs, Tests, and Operations](#) - 1,998 words
- [Allocation — Orientation and Evidence Baseline](#) - 1,992 words
- [Allocation — Ubiquitous Language, Aggregates, and State](#) - 1,991 words
- [Allocation — Commands, Events, APIs, Workflows, and Tools](#) - 1,995 words
- [Allocation — Data, Tables, Migrations, Lineage, and Reconciliation](#) - 1,995 words
- [Allocation — Portals, AI, Security, SLOs, Tests, and Operations](#) - 1,993 words

- [Pick Pack Ship — Orientation and Evidence Baseline](#) - 2,002 words
- [Pick Pack Ship — Ubiquitous Language, Aggregates, and State](#) - 2,001 words
- [Pick Pack Ship — Commands, Events, APIs, Workflows, and Tools](#) - 2,005 words
- [Pick Pack Ship — Data, Tables, Migrations, Lineage, and Reconciliation](#) - 2,005 words
- [Pick Pack Ship — Portals, AI, Security, SLOs, Tests, and Operations](#) - 2,003 words
- [Customs and Border Handling — Orientation and Evidence Baseline](#) - 2,007 words
- [Customs and Border Handling — Ubiquitous Language, Aggregates, and State](#) - 2,006 words
- [Customs and Border Handling — Commands, Events, APIs, Workflows, and Tools](#) - 2,010 words
- [Customs and Border Handling — Data, Tables, Migrations, Lineage, and Reconciliation](#) - 2,010 words
- [Customs and Border Handling — Portals, AI, Security, SLOs, Tests, and Operations](#) - 2,008 words
- [Exceptions and Claims — Orientation and Evidence Baseline](#) - 2,002 words
- [Exceptions and Claims — Ubiquitous Language, Aggregates, and State](#) - 2,001 words
- [Exceptions and Claims — Commands, Events, APIs, Workflows, and Tools](#) - 2,005 words
- [Exceptions and Claims — Data, Tables, Migrations, Lineage, and Reconciliation](#) - 2,005 words
- [Exceptions and Claims — Portals, AI, Security, SLOs, Tests, and Operations](#) - 2,003 words
- [Fulfilment and Supply Chain Domain Map](#) - 1,987 words ## 03-DETAILED-VOLUMES/volume-08-engagement-communication-architecture
- [Notification — Orientation, Ownership, Consent, and Channel Capability](#) - 1,989 words
- [Notification — Messages, Conversations, Templates, and State](#) - 1,988 words
- [Notification — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation](#) - 1,992 words
- [Notification — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests](#) - 1,995 words
- [WhatsApp — Orientation, Ownership, Consent, and Channel Capability](#) - 1,989 words
- [WhatsApp — Messages, Conversations, Templates, and State](#) - 1,988 words
- [WhatsApp — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation](#) - 1,992 words
- [WhatsApp — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests](#) - 1,995 words
- [WhatsApp Connector — Orientation, Ownership, Consent, and Channel Capability](#) - 1,993 words
- [WhatsApp Connector — Messages, Conversations, Templates, and State](#) - 1,992 words
- [WhatsApp Connector — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation](#) - 1,996 words
- [WhatsApp Connector — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests](#) - 1,999 words
- [Instagram — Orientation, Ownership, Consent, and Channel Capability](#) - 1,989 words
- [Instagram — Messages, Conversations, Templates, and State](#) - 1,988 words
- [Instagram — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation](#) - 1,992 words
- [Instagram — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests](#) - 1,995 words
- [Email — Orientation, Ownership, Consent, and Channel Capability](#) - 1,989 words
- [Email — Messages, Conversations, Templates, and State](#) - 1,988 words
- [Email — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation](#) - 1,992 words
- [Email — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests](#) - 1,995 words
- [WebSocket and Realtime Messaging — Orientation, Ownership, Consent, and Channel Capability](#) - 2,001 words
- [WebSocket and Realtime Messaging — Messages, Conversations, Templates, and State](#) - 2,000 words
- [WebSocket and Realtime Messaging — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation](#) - 2,004 words
- [WebSocket and Realtime Messaging — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests](#) - 2,007 words
- [Conversation Ownership — Orientation, Ownership, Consent, and Channel Capability](#) - 1,993 words
- [Conversation Ownership — Messages, Conversations, Templates, and State](#) - 1,992 words
- [Conversation Ownership — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation](#) - 1,996 words
- [Conversation Ownership — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests](#) - 1,999 words
- [Consent and Channel Capability — Orientation, Ownership, Consent, and Channel Capability](#) - 2,001 words
- [Consent and Channel Capability — Messages, Conversations, Templates, and State](#) - 2,000 words

- [Consent and Channel Capability — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation](#) - 2,004 words
- [Consent and Channel Capability — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests](#) - 2,007 words
- [Template and Localization — Orientation, Ownership, Consent, and Channel Capability](#) - 1,997 words
- [Template and Localization — Messages, Conversations, Templates, and State](#) - 1,996 words
- [Template and Localization — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation](#) - 2,000 words
- [Template and Localization — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests](#) - 2,003 words
- [Human Handoff and AI-Assisted Response — Orientation, Ownership, Consent, and Channel Capability](#) - 2,005 words
- [Human Handoff and AI-Assisted Response — Messages, Conversations, Templates, and State](#) - 2,004 words
- [Human Handoff and AI-Assisted Response — Provider Adapters, Webhooks, Retry, Failure, and Reconciliation](#) - 2,008 words
- [Human Handoff and AI-Assisted Response — Portals, CRM, AI Assistance, Human Handoff, SLOs, and Tests](#) - 2,011 words
- [Engagement Channel Ownership and Consent Matrix](#) - 1,987 words ## 03-DETAILED-VOLUMES/volume-09-ai-operating-system
- [AI Operating System Introduction](#) - 1,997 words
- [Platform AI Runtime SDK](#) - 1,997 words
- [AI Gateway and Model Access](#) - 1,999 words
- [AI Orchestrator Ownership](#) - 1,995 words
- [Agent Run State Machine](#) - 1,997 words
- [Supervisor and Delegation](#) - 1,995 words
- [Context Builder](#) - 1,993 words
- [Planner and Plan Versioning](#) - 1,997 words
- [Critic and Verifier](#) - 1,995 words
- [Policy and Approval Engine](#) - 1,997 words
- [Typed Tool Executor](#) - 1,995 words
- [Outcome Observer](#) - 1,993 words
- [Recovery and Replanner](#) - 1,995 words
- [MCP Tool Registry](#) - 1,995 words
- [MCP Tool Gateway](#) - 1,995 words
- [Workflow Engine Integration](#) - 1,995 words
- [Event Bus Integration](#) - 1,995 words
- [Provider Adapter Architecture](#) - 1,995 words
- [Model Registry and Routing](#) - 1,997 words
- [Prompt and Template Governance](#) - 1,997 words
- [Structured Output and Schema Validation](#) - 1,999 words
- [AI Memory](#) - 1,993 words
- [RAG and Retrieval](#) - 1,995 words
- [Knowledge Cutoff and Provenance](#) - 1,997 words
- [Digital Twin and World Model](#) - 1,999 words
- [Evaluation and Promotion](#) - 1,995 words
- [Online Monitoring and Drift](#) - 1,997 words
- [Kill Switch and Incident Containment](#) - 1,999 words
- [AI Security and Prompt Injection](#) - 1,999 words
- [Browser Agent Sandboxing](#) - 1,995 words
- [Code and Data Tool Sandboxing](#) - 1,999 words
- [AI Cost and Budget Governance](#) - 1,999 words
- [Human Handoff](#) - 1,993 words
- [AI Compliance Service](#) - 1,995 words

- [AI Vision Service](#) - 1,995 words
- [AI Voice Service](#) - 1,995 words
- [AI NLP Service](#) - 1,995 words
- [AI Automation Service](#) - 1,995 words
- [AI Edge and Federated Services](#) - 1,999 words
- [AI Marketplace](#) - 1,993 words
- [AI Observability and AIOps](#) - 1,997 words
- [Predictive Analytics and Forecast ML](#) - 1,999 words
- [Fraud Engine](#) - 1,993 words
- [Search and Ranking](#) - 1,995 words
- [Domain AI Role Catalogue](#) - 1,997 words
- [AI Canonical Ownership Matrix](#) - 1,994 words ## 03-DETAILED-VOLUMES/volume-10-data-database-reference
- [Data Architecture Introduction](#) - 1,990 words
- [Conceptual Enterprise Data Model](#) - 1,992 words
- [Domain Logical Data Models](#) - 1,992 words
- [Database-per-Service Policy](#) - 1,988 words
- [Schema Ownership](#) - 1,988 words
- [Database and Schema Registry](#) - 1,992 words
- [Table Family Documentation](#) - 1,990 words
- [Identity and Keys](#) - 1,990 words
- [Foreign Keys and Cross-Domain References](#) - 1,994 words
- [Constraints and Invariants](#) - 1,990 words
- [Indexes and Query Workloads](#) - 1,992 words
- [Effective Time and Recorded Time](#) - 1,994 words
- [Data Classification](#) - 1,988 words
- [Privacy, Purpose and Residency](#) - 2,020 words
- [Retention and Legal Hold](#) - 1,992 words
- [Data Lineage](#) - 1,988 words
- [Data Quality](#) - 1,988 words
- [Correction and Retraction](#) - 1,990 words
- [Materialized Views and Projections](#) - 1,992 words
- [Partitioning and Archival](#) - 1,990 words
- [Backup and Restore](#) - 1,990 words
- [Disaster Recovery for Data](#) - 1,992 words
- [Database Migration](#) - 1,988 words
- [Data Contract Evolution](#) - 1,990 words
- [Decommissioned Databases and Tables](#) - 1,992 words
- [Database Users and Least Privilege](#) - 1,994 words
- [Row-Level Security](#) - 1,988 words
- [Encryption and Key Management](#) - 1,992 words
- [Analytical Stores and Warehouse](#) - 1,992 words
- [Vector Stores](#) - 1,988 words
- [Search Indexes](#) - 1,988 words
- [Caching and Invalidation](#) - 1,990 words
- [CDC, Outbox and Replication](#) - 2,020 words
- [Master and Reference Data](#) - 1,992 words
- [Data Observability and Reconciliation](#) - 1,992 words

- [Database Ownership and Data Lineage Matrix](#) - 1,992 words ## 03-DETAILED-VOLUMES/volume-11-api-event-workflow-tool-contracts
- [Contract Architecture Introduction](#) - 1,992 words
- [REST and BFF Standards](#) - 1,994 words
- [gRPC Standards](#) - 1,990 words
- [OpenAPI Governance](#) - 1,990 words
- [AsyncAPI Governance](#) - 1,990 words
- [Command and Event Semantics](#) - 1,994 words
- [Event Envelope](#) - 1,990 words
- [Outbox and Inbox](#) - 1,992 words
- [Idempotency](#) - 1,988 words
- [Ordering and Concurrency](#) - 1,992 words
- [Replay and Reprocessing](#) - 1,992 words
- [Dead Letter Queues](#) - 1,992 words
- [Workflow State and Timers](#) - 1,994 words
- [Compensation and Saga Semantics](#) - 1,994 words
- [Policy Contracts](#) - 1,990 words
- [MCP Tool Schemas](#) - 1,992 words
- [Compatibility and Versioning](#) - 1,992 words
- [Consumer-Driven Contracts](#) - 1,990 words
- [Contract Testing](#) - 1,990 words
- [Webhook Contracts](#) - 1,990 words
- [External Provider Adapters](#) - 1,992 words
- [Error and Problem Details](#) - 1,994 words
- [Pagination Filtering and Sorting](#) - 1,994 words
- [Correlation Causation and Trace Context](#) - 1,996 words
- [Deprecation and Retirement](#) - 1,992 words
- [Contract Conformance Matrix](#) - 1,992 words ## 03-DETAILED-VOLUMES/volume-12-platform-architecture
- [Platform Architecture Introduction](#) - 1,991 words
- [API Gateway](#) - 1,988 words
- [Service Discovery](#) - 1,988 words
- [Configuration Service](#) - 1,988 words
- [Event Bus Service](#) - 1,991 words
- [Workflow Engine Service](#) - 1,991 words
- [MCP Layer](#) - 1,988 words
- [Observability Service](#) - 1,988 words
- [Vault and Secret Platform](#) - 1,994 words
- [Report Service](#) - 1,988 words
- [Operator Backend](#) - 1,988 words
- [WebSocket Gateway](#) - 1,988 words
- [Self-Healing Service](#) - 1,988 words
- [Cache Invalidation Subscriber](#) - 1,991 words
- [Shared Contracts and Libraries](#) - 1,994 words
- [Developer Platform](#) - 1,988 words
- [Volume 12 Conformance and Evidence Matrix](#) - 1,993 words ## 03-DETAILED-VOLUMES/volume-13-infrastructure-deployment
- [Infrastructure Introduction](#) - 1,986 words
- [Local Development](#) - 1,986 words

- [Docker Compose Profiles](#) - 1,989 words
- [Container Images](#) - 1,986 words
- [Kubernetes Architecture](#) - 1,986 words
- [Kustomize Overlays](#) - 1,986 words
- [Istio Service Mesh](#) - 1,989 words
- [Helm Packaging](#) - 1,986 words
- [Terraform Architecture](#) - 1,986 words
- [Network Segmentation](#) - 1,986 words
- [Ingress and Egress](#) - 1,989 words
- [DNS and Service Discovery](#) - 1,992 words
- [Compute and Scheduling](#) - 1,989 words
- [Persistent Storage](#) - 1,986 words
- [Regional Topology](#) - 1,986 words
- [Multi-Region and DR](#) - 1,989 words
- [Secrets and Certificates](#) - 1,989 words
- [Observability Stack](#) - 1,986 words
- [Capacity and Performance](#) - 1,989 words
- [Software Supply Chain, SBOM and Provenance](#) - 2,026 words
- [Volume 13 Conformance and Evidence Matrix](#) - 1,991 words ## 03-DETAILED-VOLUMES/volume-14-security-privacy-compliance
- [Security Architecture Introduction](#) - 1,993 words
- [Zero Trust](#) - 1,990 words
- [Authentication](#) - 1,987 words
- [Authorization and ReBAC ABAC](#) - 1,996 words
- [Tenant Isolation](#) - 1,990 words
- [Delegation Security](#) - 1,990 words
- [Secrets Management](#) - 1,990 words
- [Cryptography](#) - 1,987 words
- [Privacy and Purpose](#) - 1,993 words
- [Data Residency](#) - 1,990 words
- [Application Security](#) - 1,990 words
- [API Security](#) - 1,990 words
- [Event Security](#) - 1,990 words
- [Workflow Security](#) - 1,990 words
- [Database Security](#) - 1,990 words
- [Portal and Browser Security](#) - 1,996 words
- [Mobile Security](#) - 1,990 words
- [AI Security](#) - 1,990 words
- [Prompt Injection](#) - 1,990 words
- [Browser and Code Sandboxing](#) - 1,996 words
- [Fraud and Abuse](#) - 1,993 words
- [Audit and Non-Repudiation](#) - 1,993 words
- [Threat Modeling](#) - 1,990 words
- [Incident Response](#) - 1,990 words
- [Regulatory Evidence](#) - 1,990 words
- [Volume 14 Conformance and Evidence Matrix](#) - 1,995 words ## 03-DETAILED-VOLUMES/volume-15-engineering-handbook
- [Engineering Handbook Introduction](#) - 1,989 words

- [Repository Standards](#) - 1,986 words
- [Java 21 Standards](#) - 1,989 words
- [Spring Boot Standards](#) - 1,989 words
- [TypeScript Standards](#) - 1,986 words
- [Next.js Standards](#) - 2,017 words
- [Shared UI and Design System](#) - 1,995 words
- [Domain Modeling](#) - 1,986 words
- [API Implementation](#) - 1,986 words
- [Event Implementation](#) - 1,986 words
- [Workflow Implementation](#) - 1,986 words
- [Database Migration Standards](#) - 1,989 words
- [Testing Strategy](#) - 1,986 words
- [CI CD](#) - 1,986 words
- [Dependency Management](#) - 1,986 words
- [Feature Flags](#) - 1,986 words
- [Observability Implementation](#) - 1,986 words
- [Code Review](#) - 1,986 words
- [Architecture Tests](#) - 1,986 words
- [Definition of Done and Documentation as Code](#) - 2,001 words
- [Volume 15 Conformance and Evidence Matrix](#) - 1,991 words ## 03-DETAILED-VOLUMES/volume-16-operations-sre
- [SRE Handbook Introduction](#) - 1,989 words
- [Service Ownership](#) - 1,986 words
- [SLI SLO and Error Budgets](#) - 1,995 words
- [On-Call](#) - 1,983 words
- [Runbook Standard](#) - 1,986 words
- [Incident Command](#) - 1,986 words
- [Change Management](#) - 1,986 words
- [Release and Rollback](#) - 1,989 words
- [Backup and Restore](#) - 1,989 words
- [Recovery and Reconciliation](#) - 1,989 words
- [Capacity Management](#) - 1,986 words
- [Cost Management](#) - 1,986 words
- [Security Incident Operations](#) - 1,989 words
- [AI Incident Operations](#) - 1,989 words
- [Partner and Provider Outage Handling](#) - 1,995 words
- [Data Incident Operations](#) - 1,989 words
- [Business Continuity](#) - 1,986 words
- [Post-Incident Learning](#) - 1,986 words
- [Operational Readiness Review](#) - 1,989 words
- [Executive Operational Reporting](#) - 1,989 words
- [Volume 16 Conformance and Evidence Matrix](#) - 1,991 words ## 03-DETAILED-VOLUMES/volume-17-testing-verification-conformance
- [Testing and Conformance Introduction](#) - 1,994 words
- [Unit Testing](#) - 1,988 words
- [Property Testing](#) - 1,988 words
- [Aggregate Invariant Testing](#) - 1,991 words
- [Contract Testing](#) - 1,988 words

- [Testcontainers Integration](#) - 1,988 words
- [Database Migration Verification](#) - 1,991 words
- [Architecture Tests](#) - 1,988 words
- [Tenant Isolation Tests](#) - 1,991 words
- [Security Adversarial Tests](#) - 1,991 words
- [Event Replay Tests](#) - 1,991 words
- [Workflow Restart and Compensation Tests](#) - 1,997 words
- [AI Evaluation](#) - 1,988 words
- [Portal E2E Journeys](#) - 1,991 words
- [Accessibility Testing](#) - 1,988 words
- [Performance and Capacity Testing](#) - 1,994 words
- [Resilience and Fault Injection](#) - 1,994 words
- [Disaster Recovery Exercises](#) - 1,991 words
- [Partner Certification](#) - 1,988 words
- [Continuous Conformance](#) - 1,988 words
- [Volume 17 Conformance and Evidence Matrix](#) - 1,993 words ## 03-DETAILED-VOLUMES/volume-18-migration-legacy-evolution
- [Migration and Evolution Introduction](#) - 1,992 words
- [GTOS Core and Monolith History](#) - 1,995 words
- [Duplicate Package Migration](#) - 1,989 words
- [Strangler Patterns](#) - 1,986 words
- [Anti-Corruption Layers](#) - 1,986 words
- [API Client Migration](#) - 1,989 words
- [Database Ownership Migration](#) - 1,989 words
- [Schema Correction](#) - 1,986 words
- [Deprecated Tables and Entities](#) - 1,992 words
- [Service Extraction](#) - 1,986 words
- [Traffic Cutover](#) - 1,986 words
- [Compatibility Adapters](#) - 1,986 words
- [Dual Run and Reconciliation](#) - 1,992 words
- [Rollback and Decommissioning](#) - 1,989 words
- [Semantic Versioning and Future Evolution](#) - 1,995 words
- [Volume 18 Conformance and Evidence Matrix](#) - 1,991 words ## 03-DETAILED-VOLUMES/volume-19-implementation-profiles
- [Customer Portal Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,990 words
- [Customer Portal Profile — Architecture, Data, Security, AI, and Operations](#) - 1,994 words
- [Customer Portal Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,994 words
- [Partner Portal Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,990 words
- [Partner Portal Profile — Architecture, Data, Security, AI, and Operations](#) - 1,994 words
- [Partner Portal Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,994 words
- [Provider Portal Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,990 words
- [Provider Portal Profile — Architecture, Data, Security, AI, and Operations](#) - 1,994 words
- [Provider Portal Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,994 words
- [Operator Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,988 words
- [Operator Profile — Architecture, Data, Security, AI, and Operations](#) - 1,992 words
- [Operator Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,992 words
- [Warehouse Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,988 words
- [Warehouse Profile — Architecture, Data, Security, AI, and Operations](#) - 1,992 words

- [Warehouse Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,992 words
- [Developer Platform Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,990 words
- [Developer Platform Profile — Architecture, Data, Security, AI, and Operations](#) - 1,994 words
- [Developer Platform Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,994 words
- [AI Control Center Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,992 words
- [AI Control Center Profile — Architecture, Data, Security, AI, and Operations](#) - 1,996 words
- [AI Control Center Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,996 words
- [Single-Country Deployment Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,990 words
- [Single-Country Deployment Profile — Architecture, Data, Security, AI, and Operations](#) - 1,994 words
- [Single-Country Deployment Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,994 words
- [Multi-Jurisdiction Deployment Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,990 words
- [Multi-Jurisdiction Deployment Profile — Architecture, Data, Security, AI, and Operations](#) - 1,994 words
- [Multi-Jurisdiction Deployment Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,994 words
- [Commodity Extension Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,990 words
- [Commodity Extension Profile — Architecture, Data, Security, AI, and Operations](#) - 1,994 words
- [Commodity Extension Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,994 words
- [Industry Extension Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,990 words
- [Industry Extension Profile — Architecture, Data, Security, AI, and Operations](#) - 1,994 words
- [Industry Extension Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,994 words
- [White-Label Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,988 words
- [White-Label Profile — Architecture, Data, Security, AI, and Operations](#) - 1,992 words
- [White-Label Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,992 words
- [Offline-First Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,988 words
- [Offline-First Profile — Architecture, Data, Security, AI, and Operations](#) - 1,992 words
- [Offline-First Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,992 words
- [Edge Deployment Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,990 words
- [Edge Deployment Profile — Architecture, Data, Security, AI, and Operations](#) - 1,994 words
- [Edge Deployment Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,994 words
- [High-Assurance Regulated Profile — Profile Scope, Actors, Capabilities, and Constraints](#) - 1,990 words
- [High-Assurance Regulated Profile — Architecture, Data, Security, AI, and Operations](#) - 1,994 words
- [High-Assurance Regulated Profile — Deployment, Migration, Testing, Certification, and Evidence](#) - 1,994 words ##
- [03-DETAILED-VOLUMES/volume-20-appendices-generated-reference](#)
- [Glossary](#) - 1,991 words
- [Acronym Registry](#) - 1,993 words
- [Canon Traceability](#) - 1,993 words
- [Constitution Traceability](#) - 1,993 words
- [Capability Registry](#) - 1,993 words
- [Service Registry](#) - 1,993 words
- [Port Registry](#) - 1,993 words
- [Route Registry](#) - 1,993 words
- [Database Registry](#) - 1,993 words
- [Schema Registry](#) - 1,993 words
- [Table Dictionary](#) - 1,993 words
- [API Catalogue](#) - 1,993 words
- [Event Catalogue](#) - 1,993 words
- [Workflow Catalogue](#) - 1,993 words
- [AI Tool Catalogue](#) - 1,995 words

- [AI Model and Provider Registry](#) - 1,999 words
- [Portal Route Catalogue](#) - 1,995 words
- [ADR Index](#) - 1,993 words
- [Risk Register](#) - 1,993 words
- [Exception Register](#) - 1,993 words
- [Ownership Matrix](#) - 1,993 words
- [RACI and Decision Rights](#) - 1,997 words
- [SLO Catalogue](#) - 1,993 words
- [Runbook Index](#) - 1,993 words
- [Test Evidence Catalogue](#) - 1,995 words
- [Repository Source Index](#) - 1,995 words
- [Release Manifest](#) - 1,993 words
- [Deprecation Registry](#) - 1,993 words
- [Jurisdiction Profile Registry](#) - 1,995 words
- [Final Conformance Index](#) - 1,995 words ## 04-DELIVERY-PLAYBOOKS/adr
- [ADR-0001 — Markdown Is the Canonical Source](#) - 78 words
- [ADR-0002 — Retain Canon Terminology](#) - 71 words
- [ADR-0003 — Ontology Precedes Event Architecture](#) - 77 words
- [ADR-0004 — Separate Ontology and Epistemology](#) - 74 words
- [ADR-0005 — Domain Ownership over Shared Database](#) - 73 words
- [ADR-0006 — Recommendations Are Not Decisions](#) - 76 words
- [ADR-0007 — Intent Gates Execution](#) - 70 words
- [ADR-0008 — Append-Only Correction](#) - 72 words
- [ADR-0009 — Static Portable Publication](#) - 75 words
- [ADR-0010 — Machine-Readable Registries](#) - 73 words
- [Architecture Decision Records](#) - 11 words ## 04-DELIVERY-PLAYBOOKS/appendices
- [Architecture Quality Gates](#) - 131 words
- [Architecture Review Checklist](#) - 46 words
- [Conformance Levels](#) - 48 words
- [Implementation Evidence Catalogue](#) - 80 words
- [Normative Language](#) - 21 words
- [Release Checklist](#) - 34 words ## 04-DELIVERY-PLAYBOOKS/delivery
- [Definition of Ready](#) - 96 words
- [Vertical Slice Delivery Guide](#) - 334 words
- [Architecture-Aware Backlog Decomposition](#) - 264 words
- [Contract Testing Guide](#) - 263 words
- [Production Readiness Review](#) - 260 words
- [Cutover Command Center Guide](#) - 244 words
- [Partner Certification Guide](#) - 219 words
- [Observability Dashboard Design](#) - 243 words
- [Security Verification Guide](#) - 217 words
- [Data Pipeline Engineering Guide](#) - 211 words
- [AI Release Engineering Guide](#) - 231 words
- [Performance and Capacity Engineering](#) - 215 words
- [Post-Launch Stabilization Guide](#) - 214 words
- [Ninety-Day Domain Onboarding Roadmap](#) - 337 words
- [Architecture Design Review Facilitation](#) - 290 words

- [Command Handler Reference Flow](#) - 297 words
- [Event Consumer Reference Flow](#) - 279 words
- [Business Recovery Exercise](#) - 243 words
- [Architecture Documentation Standard](#) - 265 words
- [Architecture Debt Management](#) - 234 words
- [GTOS Team Ceremonies](#) - 230 words
- [Quality Assurance Strategy](#) - 286 words
- [GTOS Project Plan Template](#) - 281 words
- [Engineer Onboarding Guide](#) - 283 words
- [Engineering Acceptance and Handover](#) - 401 words
- [Engineering Execution Guides](#) - 17 words
- [GTOS Delivery Handbook](#) - 66 words
- [Playbook — AI Use-Case Qualification](#) - 249 words
- [Playbook — Architecture Decision Review](#) - 243 words
- [Playbook — Data Product Onboarding](#) - 248 words
- [Playbook — Domain Discovery](#) - 253 words
- [Playbook — Event Storming and State Modeling](#) - 249 words
- [Playbook — Incident Learning](#) - 245 words
- [Playbook — Migration and Cutover](#) - 251 words
- [Playbook — Partner Onboarding](#) - 250 words
- [Playbook — Release Readiness](#) - 252 words
- [Playbook — Security Threat Modeling](#) - 253 words
- [Playbook — SLO and Runbook Design](#) - 243 words
- [Playbook — Truth Ownership Workshop](#) - 242 words
- [Delivery Roles and Responsibilities](#) - 197 words
- [Stage 0 — Mandate and Scope](#) - 302 words
- [Stage 1 — Reference Alignment](#) - 300 words
- [Stage 2 — Business and Domain Design](#) - 302 words
- [Stage 3 — Contract and Data Design](#) - 302 words
- [Stage 4 — Security, AI, and Operational Design](#) - 305 words
- [Stage 5 — Build and Verification](#) - 304 words
- [Stage 6 — Controlled Launch](#) - 300 words
- [Stage 7 — Operate and Evolve](#) - 308 words
- [Work Product Catalogue](#) - 147 words ## 04-DELIVERY-PLAYBOOKS/implementation-blueprints
- [AI Agent Blueprint](#) - 249 words
- [Domain Service Blueprint](#) - 274 words
- [Federated Integration Blueprint](#) - 199 words
- [Implementation Blueprints](#) - 22 words
- [Reference Deployment Blueprint](#) - 186 words ## 04-DELIVERY-PLAYBOOKS/reference-matrices
- [Concept-to-Artifact Matrix](#) - 105 words
- [Domain Dependency Matrix](#) - 143 words
- [Reference Matrices](#) - 19 words
- [Output and Authority Matrix](#) - 87 words
- [GTOS Verification Matrix](#) - 90 words ## 04-DELIVERY-PLAYBOOKS/reference-patterns
- [Correction Propagation Pattern](#) - 132 words
- [Decision Recording Pattern](#) - 131 words
- [Event Evolution Pattern](#) - 154 words

- [Evidence Chain Pattern](#) - 137 words
- [Exception and Case Management Pattern](#) - 132 words
- [Federated Truth Pattern](#) - 144 words
- [Identity Resolution Pattern](#) - 186 words
- [GTOS Reference Patterns](#) - 22 words
- [Long-Running Trade Workflow Pattern](#) - 180 words
- [Relationship-Aware Access Pattern](#) - 124 words
- [Temporal Reconstruction Pattern](#) - 177 words ## 04-DELIVERY-PLAYBOOKS/scenarios
- [Scenario — Customs Declaration and Release](#) - 186 words
- [End-to-End Trade Scenarios](#) - 18 words
- [Scenario — Multi-Party Shipment Consolidation](#) - 148 words
- [Scenario — Pharmaceutical Cold Chain](#) - 230 words
- [Scenario — Sanctions and Restricted-Party Screening](#) - 157 words
- [Scenario — Trade Finance and Settlement](#) - 162 words ## 04-DELIVERY-PLAYBOOKS/standards
- [GTOS API Contract Standard](#) - 252 words
- [Conformance Assessment Standard](#) - 142 words
- [GTOS Event Contract Standard](#) - 233 words
- [Financial Control Standard](#) - 168 words
- [Cross-Cutting Standards](#) - 18 words
- [Jurisdiction Model](#) - 166 words
- [Semantic Migration Standard](#) - 161 words
- [End-to-End Observability Standard](#) - 149 words
- [GTOS Policy Standard](#) - 215 words
- [Privacy and Purpose Standard](#) - 178 words ## 04-DELIVERY-PLAYBOOKS/visual-atlas
- [GTOS Visual Architecture Atlas](#) - 77 words ## 08-CURRENT-PROJECT-GAP-ANALYSIS/GTOS-Handbook-Batch-07-Audit-Canonicalization-Evidence-Wave-01
- [Conversation Intent Review and Access Boundary](#) - 258 words
- [Final Artifact Inventory and Integrity Review](#) - 180 words
- [Content Quality Audit](#) - 180 words
- [Claim and Evidence Audit](#) - 155 words
- [Structure, Numbering, and Canonicalization Audit](#) - 130 words
- [PDF, XLSX, and ZIP Audit](#) - 178 words
- [Critical Findings and Issue Register](#) - 201 words
- [Remediation Decisions](#) - 133 words
- [Repository Evidence Wave 01 - Proforma Invoice](#) - 311 words
- [Repository Evidence Wave 01 - Payment](#) - 322 words
- [Repository Evidence Wave 01 - Purchase Order](#) - 351 words
- [Repository Evidence Wave 01 - Manufacturing](#) - 302 words
- [Evidence Wave 01 Traceability Matrix](#) - 147 words
- [Next Repository Evidence Wave](#) - 83 words
- [Batch 07 Integrity Report](#) - 25 words
- [GTOS Handbook - Batch 07](#) - 86 words ## 08-CURRENT-PROJECT-GAP-ANALYSIS/GTOS-Handbook-Batch-08-Repository-Evidence-Wave-02
- [00 — Scope, Method, and Repository Snapshot](#) - 209 words
- [01 — Quality Inspection, QC Hold, and Release](#) - 469 words
- [02 — Packing List and Commercial Invoice](#) - 363 words
- [03 — Export Customs and Legal Release](#) - 274 words
- [04 — Import Customs, Duties, and Release](#) - 263 words

- [05 — Transport Booking, Handover, Shipment, and Tracking](#) - 366 words
- [06 — Warehouse Receiving and Receipt](#) - 275 words
- [07 — Inventory Synchronization, Reservation, and QC Disposition](#) - 318 words
- [08 — Cross-Domain Contradictions and Canonicalization Register](#) - 210 words
- [09 — Implementation Status Matrix](#) - 172 words
- [10 — Wave 02 Issue Register](#) - 492 words
- [11 — Wave 02 Traceability Matrix](#) - 370 words
- [12 — Remediation and Verification Plan](#) - 226 words
- [13 — Repository Evidence Wave 03 Plan](#) - 84 words
- [GTOS Batch 08 — Repository Evidence Wave 02 Combined Edition](#) - 4,461 words
- [Implementation and Evidence Status Vocabulary](#) - 105 words
- [Integrity Report](#) - 79 words
- [GTOS Handbook — Batch 08](#) - 105 words
- [Batch 08 Release Report](#) - 100 words ## 08-CURRENT-PROJECT-GAP-ANALYSIS/GTOS-Handbook-Batch-09-Repository-Evidence-Wave-03
- [00 — Scope, Method, and Snapshot](#) - 193 words
- [01 — Product Catalog, Marketplace Catalog, and Offer](#) - 312 words
- [02 — Pricing, Price Version, and Commercial Snapshot](#) - 223 words
- [03 — Cart, Checkout, and AI Insights](#) - 333 words
- [04 — Customer Order and Workflow](#) - 282 words
- [05 — Allocation, Fulfilment, Pick, Pack, and Last Mile](#) - 305 words
- [06 — Returns, Inspection, Disputes, and Refund](#) - 262 words
- [07 — Settlement, Reconciliation, and Payout](#) - 262 words
- [08 — ESG, Sustainability Goals, and Assurance](#) - 216 words
- [09 — AI Evaluation and Governed Learning](#) - 250 words
- [10 — Cross-Domain Conflicts and Canonicalization](#) - 258 words
- [11 — Implementation Status Matrix](#) - 178 words
- [12 — Wave 03 Issue Register](#) - 1,055 words
- [13 — Wave 03 Traceability Matrix](#) - 589 words
- [14 — Remediation and Verification Plan](#) - 268 words
- [15 — Repository Evidence Wave 04 Plan](#) - 96 words
- [GTOS Batch 09 — Repository Evidence Wave 03 Combined Edition](#) - 5,486 words
- [Implementation and Evidence Status Vocabulary](#) - 101 words
- [Integrity Report](#) - 82 words
- [GTOS Handbook — Batch 09](#) - 95 words
- [Batch 09 Release Report](#) - 142 words

Publication Statement

This edition consolidates the complete handbook corpus produced during the GTOS architecture program into one governed publication.

What this edition is

- the target enterprise architecture for GTOS;
- the domain and lifecycle specification for global trade;
- the implementation guide for backend, frontend, data, events, workflows, AI, security, testing, infrastructure, and operations;
- the conformance baseline for engineering reviews and release readiness;
- the roadmap for completing and correcting the current project.

What this edition is not

- a statement that the current source code already conforms;
- legal advice or a substitute for jurisdiction-specific counsel;
- proof that a deployment is secure, compliant, resilient, or production ready;
- permission to bypass architecture, security, financial, quality, customs, or privacy approvals.

Authority hierarchy

1. Canon and Constitution.
2. Normative Core.
3. Detailed Volumes.
4. Architecture Decision Records.
5. Domain implementation specifications.
6. Code, configuration, migrations, tests, and runtime evidence.
7. Current-project gap reports.

A lower layer may implement or specialize a higher layer; it may not silently contradict it.

Final-edition interpretation

“Final” means the publication is a complete implementation blueprint and governed reference edition. It does not mean architecture can never evolve. Changes require an ADR, compatibility analysis, migration plan, conformance update, and release note.

Reading Guide

Executive and product leaders

Read the System Blueprint, Target Operating Model, Domain Map, Lifecycle, AI Governance, Security, Roadmap, and Completion Definition.

Architects

Read the Canon, Constitution, Normative Core, Domain Architecture volumes, Data and Contract volumes, Platform, Infrastructure, Security, Migration, and Conformance volumes.

Backend engineers

Start with Service Architecture, Data Architecture, API and Event Architecture, Workflow and Saga Architecture, Engineering Standards, Testing, and the relevant Domain packs.

Frontend engineers

Read Portal and Experience Architecture, Identity and Authorization, API Contracts, Event/Reactivity patterns, Accessibility, localization, observability, and the relevant Domain journeys.

Data and AI engineers

Read the Reference Theory, Data and Database Reference, AI Operating System, AI Security, Evaluation, lineage, evidence, uncertainty, and model/tool governance.

SRE and security engineers

Read Platform, Infrastructure, Security, Operations, SRE, Testing, Disaster Recovery, Incident Management, and Production Readiness.

How to implement a feature

For every feature, traverse this chain:

Business capability
-> Domain owner
-> Aggregate and invariant
-> Decision and authority
-> Command and workflow
-> API and Event contract
-> Database ownership
-> Portal journey
-> Security and privacy controls
-> SLO and runbook
-> Tests and release evidence

GTOS Enterprise Engineering Handbook

Final Edition 1.0

Global Trade Operating System - Enterprise Architecture, Domain Design, Implementation, Security, AI, Data, Delivery, and Operations Handbook

This package is the normative design and implementation reference for building GTOS as a global, multi-tenant, event-driven, AI-native trade operating system.

The main book describes **what GTOS shall become and how it shall be implemented**. The current repository is not treated as the architecture authority. Repository findings are isolated under 08-CURRENT-PROJECT-GAP-ANALYSIS and may be used only to plan migration toward the target architecture.

Start here

1. Read PUBLICATION-STATEMENT.md.
2. Read MASTER-TABLE-OF-CONTENTS.md.
3. Read the documents under 01-NORMATIVE-CORE in numeric order.
4. Use 03-DETAILED-VOLUMES as the comprehensive domain and platform reference.
5. Use 04-DELIVERY-PLAYBOOKS to implement each vertical slice.
6. Use 06-CATALOGUES and 09-BUILD-AND-VALIDATION for navigation and conformance.

Edition rule

The words **shall**, **must**, and **required** are normative. **Should** identifies the default engineering choice unless an Architecture Decision Record authorizes an exception. **May** identifies an allowed option.

Primary outcome

A team conforming to this handbook shall be able to design, implement, verify, deploy, operate, recover, audit, and evolve GTOS without relying on undocumented tribal knowledge.

Release Notes - Final Edition 1.0

Consolidated

- Canon, Constitution, reference theory, vision, business operating model.
- End-to-end 27-stage global trade lifecycle.
- Portal, identity, tenant, and organization architecture.
- Commerce, fulfilment, supply-chain, engagement, and communications domains.
- AI operating system, data, APIs, Events, workflows, and tools.
- Platform, infrastructure, security, privacy, engineering, SRE, testing, and migration.
- Implementation profiles, generated references, delivery playbooks, diagrams, catalogues, and current-project gap appendices.

Editorial changes

- Target architecture is now the main publication.
- Current repository issues are isolated in an appendix.
- A normative core and completion definition were added.
- A single master table of contents and artifact manifest were generated.
- PDF books are split into coherent volumes for reliable reading and printing.

Version

- Product: GTOS Enterprise Engineering Handbook
- Edition: 1.0 Final Master Edition
- Publication date: 2026-07-23
- Packaging format: Markdown source, PDF collection, diagrams, catalogues, playbooks, validation artifacts
- Primary language: English technical reference, with internationalization requirements for implementation
- Change model: semantic editioning plus ADR-governed amendments

GTOS System Blueprint

Mission

GTOS is a global trade operating system that coordinates commercial intent, organizations, suppliers, products, procurement, contracts, trade documents, payments, production, quality, customs, transport, warehouses, inventory, market-places, customer orders, delivery, returns, settlement, reporting, ESG, and AI-assisted operations.

GTOS shall not be designed as a collection of screens around a shared database. It shall be designed as a network of accountable bounded contexts that exchange typed contracts and preserve independent truth ownership.

Architectural shape

GTOS uses five interacting planes:

1. **Business Domain Plane** - canonical aggregates, Decisions, obligations, Commands, Events, and history.
2. **Experience Plane** - customer, partner, provider, operator, warehouse, admin, developer, and AI-control Portals.
3. **Integration Plane** - APIs, Event Bus, workflow engine, provider adapters, webhooks, files, EDI, and MCP tools.
4. **Intelligence Plane** - context assembly, retrieval, planning, recommendations, tool execution, evaluation, and governed autonomy.
5. **Platform Plane** - identity, policy, secrets, observability, deployment, data infrastructure, reliability, and developer platform.

Foundational truth model

GTOS shall preserve the following distinctions:

Entity != Observation != Claim != Evidence
Prediction != Recommendation != Decision
Decision != Intent != Command != Execution != Outcome
Custody != Ownership
Application status != external legal authority
Document != truth
Projection != source of record

Every material fact must identify its owner, subject, source, time, effective interval, confidence or qualification, policy context, authority, and correction lineage.

System boundaries

GTOS coordinates external systems but does not own their authority. Banks own provider payment outcomes. Customs authorities own legal release. Carriers own qualified carrier observations. Laboratories and inspectors own their signed observations. The platform records, verifies, relates, and derives perspectives from those outcomes.

Reference deployment

- multi-region capable;
- tenant- and jurisdiction-aware;
- database-per-service or strongly isolated schema ownership;
- asynchronous integration through durable Events;
- durable workflow for long-running trade processes;
- zero-trust service identity;
- typed AI tools rather than direct model access to databases;
- immutable audit for binding Decisions and privileged execution;
- observable and recoverable partial failure.

Non-negotiable qualities

- historical integrity;
- explicit authority;
- deterministic identifiers;
- idempotent execution;
- secure tenant isolation;
- correction without erasure;
- explainable automation;
- contract compatibility;
- operational recovery;
- evidence-backed conformance.

Architectural Principles

1. Truth has an owner

Every proposition has exactly one authoritative owner at a given level of abstraction. Other services may cache or project it but may not mutate it by shared-table access.

2. History is append-oriented

Approved, issued, submitted, signed, paid, released, inspected, delivered, and settled facts shall not be rewritten. Corrections and revocations create new facts linked to the originals.

3. Authority is evaluated at action time

Authentication alone is insufficient. Every binding action must evaluate principal, represented organization, role, relationship, delegation, mandate, jurisdiction, purpose, and policy.

4. Distributed execution assumes partial failure

Every external or cross-service action must support idempotency, timeouts, retries, duplicate messages, out-of-order delivery, replay, outcome unknown, reconciliation, and compensation.

5. Events describe completed facts

Commands use imperative names. Events use past-tense completed-fact names. Plans, requests, and predictions are not Events.

6. AI is governed software

AI may extract, classify, summarize, predict, recommend, plan, or invoke typed tools within policy. It shall not own canonical truth, invent authority, conceal uncertainty, or bypass approval.

7. Security is architectural

Tenant isolation, least privilege, data minimization, secret isolation, integrity, audit, and incident containment are designed into every Domain and contract.

8. Operational ownership is part of design

Every service and workflow has an owner, SLO, dashboard, alert, runbook, capacity plan, dependency map, recovery procedure, and decommissioning plan.

9. Compatibility is explicit

APIs, Events, database migrations, files, tools, prompts, and policy versions evolve through documented compatibility rules.

10. Completion requires evidence

A capability is not complete because code compiles. It is complete when invariants, security, interoperability, recovery, performance, and business outcomes are proven by executable evidence.

Target Operating Model

Product organization

GTOS should be operated through durable product and platform teams aligned to bounded contexts, not temporary component projects.

Business product groups

- Identity and Organization;
- Customer and CRM;
- Product, Catalog, Marketplace, Pricing, and Promotions;
- Procurement, Quotation, Contract, PI, and Purchase Order;
- Payment, Wallet, Billing, Finance Control, and Settlement;
- Manufacturing and Quality;
- Shipping, Customs, Provider, and Regulatory;
- Warehouse, Inventory, Fulfilment, Delivery, and Returns;
- Engagement and Communications.

Platform groups

- Developer Platform and API Gateway;
- Data Platform;
- Event and Workflow Platform;
- Identity and Security Platform;
- AI Platform and Model Governance;
- Observability and Reliability Platform;
- Cloud and Infrastructure Platform.

Decision rights

- Domain teams own business semantics and invariants.
- Platform teams own reusable technical capabilities and paved roads.
- Security and privacy authorities define mandatory controls and exceptions.
- Architecture governance resolves ownership and compatibility disputes.
- SRE defines production readiness and reliability policy.
- AI Governance approves autonomy, tools, evaluation, and kill-switch policy.

Delivery model

Work is delivered as vertical slices that include Domain behavior, persistence, contract, Portal journey, security, observability, tests, migration, runbook, and release evidence.

Governance cadence

- weekly Domain design review;
- biweekly contract and integration review;
- monthly architecture and security council;
- quarterly resilience, disaster recovery, and AI governance review;
- release-specific production readiness review;
- annual Canon and Constitution review.

Domain Map and Ownership

Core commercial domains

- **Product Catalog** owns canonical product identity and versioned product content.
- **Marketplace** owns seller/channel offers and publication eligibility.
- **Pricing** owns Price Versions and pricing policies.
- **Cart and Checkout** own shopping intent and authoritative checkout quotes.
- **Order** owns accepted customer commercial obligations.
- **Payment** owns Payment Intent and provider financial outcomes.
- **Settlement** owns financial entitlements, reconciliation, payout Intent, and settlement state.

Trade execution domains

- **Procurement** owns RFQ, supplier invitations, and Purchase Orders.
- **Quotation** owns supplier commercial proposals and versions.
- **Contract** owns negotiated binding terms.
- **Trade Documents** owns Proforma Invoice, Packing List, Commercial Invoice, certificates, and document versions.
- **Manufacturing** owns production plans, orders, lots, progress, and capacity commitments.
- **Quality** owns inspection, measurements, nonconformance, and quality-release Decisions.
- **Regulatory and Customs** own platform declarations and authority observations.
- **Shipping** owns shipment, legs, booking relationships, transport observations, and exceptions.

Fulfilment domains

- **Warehouse** owns physical receiving, storage operations, pick, pack, and dispatch handover.
- **Inventory** owns stock positions, reservations, allocations, movements, and adjustments.
- **Fulfilment** coordinates accepted demand to physical execution without taking ownership of Order or Inventory truth.
- **Delivery** owns last-mile attempts and qualified Proof of Delivery observations.
- **Returns** owns return authorization, reverse logistics, inspection, disposition, claims, and disputes.

Supporting domains

- Identity, Organization, Tenant, Customer, Supplier, Provider, CRM, Communications, Reporting, ESG, and AI Evaluation.

Ownership rule

A shared concept may appear in several contexts as a reference or projection. Only one context owns each authoritative proposition. Ownership is recorded in the Domain Catalogue and enforced by contracts and database boundaries.

End-to-End Global Trade Lifecycle

The canonical lifecycle contains 27 stages:

1. Organization identity and onboarding.
2. Demand and requirement definition.
3. Request for quotation.
4. Supplier discovery and qualification.
5. Quotation invitation and submission.
6. Comparison and normalization.
7. Negotiation and supplier selection.
8. Contract and accepted terms.
9. Proforma Invoice.
10. Payment and risk approval.
11. Purchase Order and acknowledgement.
12. Production planning and capacity commitment.
13. Production execution and milestones.
14. Quality inspection, nonconformance, and release.
15. Packing List and Commercial Invoice.
16. Export customs and legal release.
17. Transport planning and booking.
18. Pickup, custody, and handover.
19. International transport and tracking.
20. Import customs, duties, and legal release.
21. Delivery and warehouse receiving.
22. Inventory synchronization and availability.
23. Product publication and marketplace eligibility.
24. Checkout and customer order.
25. Allocation, fulfilment, and last-mile delivery.
26. Returns, claims, refunds, and disputes.
27. Settlement, reporting, ESG, and learning.

Stage contract

Every stage must specify:

- owner and external authority;
- entry and exit criteria;
- canonical subjects and versions;
- state machines and invariants;
- Decisions, Commands, Events, and queries;
- evidence and documents;
- workflow, deadlines, retries, and compensation;
- APIs, database ownership, and Portal journeys;
- AI participation and autonomy;
- security, privacy, SLOs, tests, and recovery;
- correction and dispute behavior.

Lifecycle orchestration

The lifecycle is not a single central transaction. Durable workflows coordinate independently owned aggregates. State is advanced only by verified facts. If an external outcome is unknown, the process enters an explicit reconciliation state rather than assuming success or repeating the action blindly.

Service Architecture

Service boundary

A service is justified by independent business ownership, lifecycle, scaling, security, data, or operational needs. A table group or UI menu is not sufficient reason for a service.

Each service contains:

- API / Adapter Layer
- Application Layer
- Domain Layer
- Persistence and Integration Layer
- Security and Policy Layer
- Event and Workflow Layer
- Observability and Operations Layer
- Test and Conformance Layer

Database ownership

Each service owns its writes. Cross-service database joins and updates are prohibited. Read models may be built through Events, APIs, CDC under governance, or analytical pipelines.

Synchronous communication

Use synchronous calls for bounded request-response interactions where the caller requires an immediate answer and the dependency can meet the latency and availability budget. Apply timeouts, circuit breaking, authentication, authorization, tracing, idempotency, and explicit error contracts.

Asynchronous communication

Use Events for completed facts and durable decoupling. Producers commit state plus outbox atomically. Consumers use inbox/idempotency, versioned schemas, replay safety, ordering strategy, DLQ, and reconciliation.

Long-running processes

Use a durable workflow engine or persisted saga state for multi-step trade processes. Workflow state is coordination state, not Domain truth.

Service template

Every service repository module shall include ownership metadata, ADRs, API/Event specifications, migrations, dependency policy, SLO, dashboards, alerts, runbooks, threat model, test strategy, and deprecation plan.

Data Architecture

Principles

- database ownership follows Domain ownership;
- identifiers are stable, opaque, and namespace-aware;
- recorded time and effective time are distinct;
- binding versions are immutable;
- corrections append rather than overwrite;
- lineage is first-class;
- sensitive data is classified and minimized;
- analytical and AI stores are projections.

Storage patterns

- relational databases for transactional aggregates and constraints;
- object storage for documents, evidence, large media, and signed artifacts;
- search indexes for discovery and text retrieval;
- time-series systems for metrics and telemetry;
- graph or relationship projections where relationship traversal is central;
- vector indexes for governed retrieval, never as source of truth;
- warehouse/lakehouse for analytical and regulatory reporting.

Schema design

Every table has tenant or ownership scope where applicable, primary key, business-key uniqueness, version or concurrency mechanism, timestamps, classification, and retention policy. Monetary values store amount plus currency. Physical measures store value plus unit. External references store issuer and namespace.

Migration standard

Migrations are forward-only, reviewable, repeatable in CI, compatible with rolling deployment, and tested against empty and realistic upgrade datasets. Destructive changes require export, backfill, validation, cutover, rollback, and retention approval.

Recovery

Each datastore defines RPO, RTO, backup schedule, encryption, restore test, regional strategy, corruption detection, and reconciliation process.

API, Event, and Tool Architecture

REST and BFF

APIs use resource or command-oriented contracts with explicit versions, idempotency, optimistic concurrency, pagination, filtering, problem details, authorization, and trace propagation. Tenant identity is derived from trusted authentication context, never from an untrusted browser header.

Events

Every Event includes event ID, type, schema version, subject, tenant, producer, occurred time, recorded time, correlation, causation, trace context, source version, and payload.

Events are immutable completed facts. Consumers must tolerate duplicates and unknown fields. Breaking semantic changes require a new Event type or governed version migration.

Webhooks and external callbacks

Verify provider identity, signature, timestamp, nonce/replay protection, payload schema, and correlation. Store the original message. Respond quickly, process asynchronously, and reconcile missing or contradictory outcomes.

Files and EDI

File exchanges define format, encoding, naming, checksums, signatures, encryption, partial-file handling, duplicate detection, acknowledgement, retention, and reconciliation.

AI and MCP tools

Tools expose narrow typed operations with schemas, policy, authorization, approval, idempotency, timeout, cost limit, audit, and outcome observation. Models never receive direct database credentials.

Workflow, Saga, and Recovery Architecture

Workflow responsibilities

A workflow coordinates time, dependencies, approvals, external calls, retries, and compensation. It does not own the business facts produced by Domain services.

Required workflow state

- process ID and business correlation;
- participating subjects and exact versions;
- current step and deadlines;
- submitted Commands and idempotency keys;
- observed Events and external responses;
- approvals and authority;
- retry budget and next action;
- compensation status;
- operator interventions;
- final outcome and audit.

Outcome Unknown

When a call times out after possible external execution, the workflow records Outcome Unknown. It queries the provider, waits for a verified callback, or opens reconciliation. It does not retry a financial, customs, custody, or irreversible physical action blindly.

Compensation

Compensation creates a new authorized Intent: release reservation, cancel booking, reverse ledger entry, issue correction, or request refund. It never deletes the original completed fact.

Operational requirements

Workflows must survive process restart, deployment, broker outage, duplicate Event, late Event, timer loss, and manual intervention. Stalled and orphaned instances are observable and recoverable.

AI-Native Architecture

AI control plane

GTOS AI consists of:

- AI Gateway and provider abstraction;
- Model Router and policy-aware selection;
- Context Builder and governed retrieval;
- Prompt and instruction registry;
- Agent Runtime and state machine;
- Planner, critic, verifier, and supervisor;
- Tool Registry and execution gateway;
- short-term, episodic, semantic, and organizational memory;
- evaluation, observability, safety, incident, and kill-switch systems.

Autonomy levels

- A0: information only;
- A1: recommendation requiring human Decision;
- A2: automatic low-risk action with reviewable evidence;
- A3: bounded workflow autonomy under policy and approval thresholds;
- A4: exceptional high autonomy allowed only for explicitly certified domains.

Financial, legal, customs, quality release, custody, privileged security, and authoritative correction actions default to A0 or A1.

Agent run record

Every run records tenant, principal, purpose, input, knowledge cutoff, context sources, prompt/model/tool versions, policy evaluations, approvals, tool calls, structured outputs, uncertainty, costs, observed outcomes, corrections, and evaluation.

RAG and memory

Retrieved text is evidence, not authority. Sources require access control, version, effective time, provenance, and freshness. Memory cannot silently create canonical business facts.

Evaluation

Evaluate quality, factuality, safety, security, policy compliance, tool correctness, business outcome, latency, cost, drift, and subgroup performance. Promotions require sample thresholds, human approval, rollback, and incident controls.

Security, Privacy, and Compliance

Identity

GTOS recognizes human, organization, service, workload, device, provider, and AI Agent identities. Authentication uses standards-based federation, strong service identity, MFA for privileged users, and managed session lifecycles.

Authorization

Authorization evaluates principal, represented party, tenant, organization, relationship, role, attributes, delegation, mandate, purpose, resource, action, jurisdiction, risk, and time. High-impact actions require separation of duties and step-up authentication.

Tenant isolation

Tenant context is established at the trusted edge, cryptographically or operationally bound to identity, propagated through calls and Events, and enforced in every repository query, cache key, search index, object path, log, metric, and analytical dataset. Tests attempt cross-tenant access continuously.

Data protection

Classify data, minimize collection, encrypt in transit and at rest, tokenize high-risk values, isolate keys and secrets, rotate credentials, prevent sensitive logs/prompts, and enforce purpose, consent, retention, residency, legal hold, export, and deletion rules.

Application and integration security

Apply threat modeling, secure coding, dependency and container scanning, signed artifacts, SBOM, rate limits, abuse detection, webhook verification, SSRF protection, file scanning, browser sandboxing, prompt-injection defenses, and privileged tool controls.

Audit and non-repudiation

Binding Decisions, document issuance, payments, customs submissions, quality release, custody, settlement, access-policy changes, AI tool execution, and emergency access require immutable audit with actor, authority, source, time, reason, before/after, and integrity evidence.

Reliability, SRE, and Operations

Reliability model

Availability is only one dimension. GTOS defines durability, correctness, freshness, latency, recoverability, security, and business deadline objectives.

Service level management

Each user journey and service defines SLIs, SLOs, error budgets, dependencies, capacity, failure modes, and escalation. Business process SLOs track stuck orders, expiring documents, customs holds, delayed shipments, unprocessed returns, and unreconciled payments.

Observability

Use structured logs, metrics, traces, profiles, Event-flow telemetry, workflow state, audit links, business KPIs, and synthetic journeys. Correlation IDs connect Portal action through workflow, service, database, Event, provider, and final outcome.

Incident management

Define severity, command roles, communication, evidence preservation, containment, recovery, customer/regulatory notification, reconciliation, correction, and post-incident learning.

Disaster recovery

Test backups, restore, regional failover, broker recovery, workflow replay, secret/key recovery, object-store recovery, external-provider reconciliation, and data-integrity verification. A DR document without a successful exercise is not evidence.

Change and release

Use progressive delivery, feature flags, compatibility windows, migration gates, automated rollback criteria, release observability, change freeze rules, and post-launch stabilization.

Engineering Standards

Code structure

Use clear Domain, application, adapter, and infrastructure boundaries. Domain code must not depend on transport frameworks. Persistence models should not dictate business semantics.

Quality gates

Every change passes formatting, static analysis, dependency rules, unit/property tests, integration tests, contract tests, security checks, migration checks, build reproducibility, and documentation validation.

Git and review

Use small reviewable changes, protected branches, signed or attributable commits, mandatory review for critical areas, CODEOWNERS, semantic commit messages, and linked ADR/issue/test evidence.

Configuration

Configuration is typed, validated, environment-specific, secret-free in source, and observable. Defaults must be safe. Runtime changes are audited.

Error handling

Errors are typed and classified as validation, authorization, conflict, dependency, timeout, outcome unknown, business rejection, or internal defect. Do not expose secrets or implementation details.

Documentation as code

Architecture, APIs, Events, migrations, operational procedures, diagrams, and conformance evidence live with version control and CI validation.

Testing and Conformance

Test layers

- unit tests for pure rules and invariants;
- property-based tests for state and numerical constraints;
- database/migration tests with real engines;
- component integration tests with real dependencies;
- API and Event contract tests;
- workflow restart, timeout, duplicate, ordering, and compensation tests;
- security, authorization, tenant-isolation, privacy, and abuse tests;
- Portal E2E tests;
- performance, capacity, soak, and resilience tests;
- backup/restore and disaster-recovery exercises;
- AI evaluation and adversarial tests.

Conformance levels

- C0: documentation skeleton;
- C1: Domain semantics and contracts defined;
- C2: implementation and automated tests exist;
- C3: security, observability, recovery, and migration verified;
- C4: production evidence and operational ownership established;
- C5: independently certified for high-assurance use where required.

Release evidence

A release package contains versioned artifacts, SBOM, signatures, migration report, test results, security findings, compatibility report, performance evidence, runbook, rollback plan, dashboards, approvals, and known limitations.

Implementation Roadmap

Phase 0 - Governance and foundations

Adopt Canon, Constitution, Domain ownership, identity model, tenant model, engineering standards, ADR process, catalogues, CI quality gates, and production-readiness policy.

Phase 1 - Platform paved roads

Deliver identity, service templates, API gateway, secrets, observability, Event Bus, outbox/inbox, workflow engine, database migration framework, developer environment, test harnesses, and deployment platform.

Phase 2 - Core commerce

Implement Product, Marketplace Offer, Price Version, Cart, Checkout Quote, Customer Order, Payment Intent, Inventory Reservation, and reliable Order saga.

Phase 3 - Trade and supply chain

Implement RFQ, Quotation, Contract, PI, PO, Manufacturing, Quality, documents, customs, shipping, custody, warehouse receiving, inventory, and fulfilment.

Phase 4 - After-sales and finance

Implement delivery evidence, returns, claims, refund, settlement, reconciliation, reporting, and ESG evidence.

Phase 5 - AI-native operations

Introduce retrieval, recommendations, planning, typed tools, supervised agents, evaluation, safety, and bounded autonomy after the underlying Domain controls are mature.

Phase 6 - Global scale and certification

Add multi-region, jurisdiction profiles, advanced resilience, partner certification, data residency, high-assurance controls, cost optimization, and formal compliance evidence.

Prioritization rule

Build the smallest complete vertical slice, not isolated tables or screens. Platform foundations must be delivered just ahead of the slices that require them, avoiding both premature platform construction and repeated local reinvention.

Domain Implementation Template

Every Domain pack shall include:

1. Purpose, scope, owner, external authorities, and non-goals.
2. Ubiquitous language and subject identities.
3. Context map and upstream/downstream relationships.
4. Aggregates, entities, value objects, invariants, and state machines.
5. Decisions, authority, policies, exceptions, appeals, and corrections.
6. Commands, Events, queries, API, webhook, file, and tool contracts.
7. Durable workflows, deadlines, retries, idempotency, reconciliation, and compensation.
8. Database schema, keys, constraints, indexes, versions, migrations, lineage, and retention.
9. Portal journeys, personas, accessibility, localization, and error states.
10. AI use cases, autonomy, context, tools, evaluation, and prohibited behavior.
11. Security, privacy, threat model, tenant isolation, and audit.
12. SLOs, dashboards, alerts, runbooks, capacity, backup, and recovery.
13. Tests, conformance evidence, rollout, migration, compatibility, and deprecation.
14. Worked scenarios including success, partial failure, contradiction, correction, and dispute.

A Domain review is rejected if any section is replaced by “TBD” without an owner, due date, and risk decision.

Production Readiness Standard

A capability may enter production only when:

- ownership and on-call are assigned;
- threat model and privacy review are complete;
- authentication, authorization, tenant isolation, rate limiting, and audit are tested;
- migrations are upgrade-tested and rollback/backout is defined;
- idempotency and concurrency behavior are proven;
- Events and workflows survive duplicate, loss, delay, restart, and replay;
- dependencies have timeouts, circuit breakers, and fallback/reconciliation;
- SLOs, dashboards, alerts, runbooks, and capacity exist;
- backup and restore have been exercised;
- performance and failure tests meet thresholds;
- Portal E2E and accessibility pass;
- AI evaluations and kill switches pass where AI is used;
- release, rollback, and communication plans are approved;
- known risks and exceptions are recorded with expiry.

Production readiness is a joint Decision by the Domain owner, platform owner, security, SRE, and relevant business authority.